National University of Lesotho

Department of Mathematics and Computer Science
M3503 – Ordinary Differential Equations

Tutorial 3 – Variation of Parameters, and the Inverse Operator Method

(September 14, 2026)

1. Variation of parameters

The method, in one line. For $y'' + P(x)y' + Q(x)y = f(x)$ with fundamental set $\{y_1, y_2\}$ and Wronskian $W = y_1y_2' - y_2y_1'$,

$$u_1' = -\frac{y_2 f}{W}, \qquad u_2' = \frac{y_1 f}{W}, \qquad y_p = u_1y_1 + u_2y_2.$$

Put the equation in *standard form* (leading coefficient 1) before reading off f. Absorb any term of y_p that duplicates y_c into the arbitrary constants.

1. Find a particular solution by variation of parameters, then state the general solution.
   (a) $y'' + y = \tan x$
   (b) $y'' + y = \csc x$
   (c) $y'' + 4y = \sec 2x$
   (d) $y'' - 2y' + y = \dfrac{e^x}{x}$
2. Solve completely.
   (a) $y'' - y = e^{-x}$. (Note how variation of parameters absorbs the resonance automatically — no separate modification rule is needed.)
   (b) $y'' - 3y' + 2y = e^{3x}$. Then **verify** your y_p by the method of undetermined coefficients.

2. The inverse operator method

Make y the subject. Write the equation as $\phi(\mathrm{D})y = f(x)$ and take $y_p = \dfrac{1}{\phi(\mathrm{D})}f(x)$. Then act by the case matching f: e^{ax}: $\mathrm{D} \to a$ ($x^r/r!$ if $\phi(a) = 0$); $\sin ax/\cos ax$: $\mathrm{D}^2 \to -a^2$; x^m: expand $[\phi(\mathrm{D})]^{-1}$ up to D^m; $e^{ax}V$: shift $e^{ax}\frac{1}{\phi(\mathrm{D}+a)}V$; xV: $\left[x - \frac{\phi'(\mathrm{D})}{\phi(\mathrm{D})}\right]\frac{1}{\phi(\mathrm{D})}V$; otherwise partial fractions with $\frac{1}{\mathrm{D}-a}f = e^{ax}\int e^{-ax}f\,dx$.

5. **Cases 1–2 (exponential and sinusoidal input).** Test $\phi(a)$ or $\phi(-a^2)$ for resonance first, then find the particular integral.
   (a) $(\mathrm{D}^2 - 5\mathrm{D} + 6)\,y = e^{4x}$
   (b) $(\mathrm{D}^2 + 3\mathrm{D} - 4)\,y = e^{-4x}$
   (c) $(\mathrm{D}^2 + 2\mathrm{D} + 5)\,y = \sin x$
   (d) $(\mathrm{D}^2 + 16)\,y = \cos 4x$

6. **Case 3 (polynomial input).** Find the particular integral.
   (a) $(\mathrm{D}^2 + \mathrm{D} + 1)\, y = x^2$
   (b) $(\mathrm{D}^2 - 4)\, y = x^2 + 1$
   (c) ⋆ $(\mathrm{D}^2 + 2\mathrm{D})\, y = x$. The constant term of $\phi(\mathrm{D})$ is zero here — factor out a D and integrate last. Why can you not expand $[\phi(\mathrm{D})]^{-1}$ directly?

7. **Cases 4–5 (shift theorem; input $x\,V(x)$).** Find the particular integral.
   (a) $(\mathrm{D}^2 - 2\mathrm{D} + 1)\, y = x\,\mathrm{e}^{x}$
   (b) $(\mathrm{D}^2 + 4\mathrm{D} + 4)\, y = x^2\,\mathrm{e}^{-2x}$
   (c) $(\mathrm{D}^2 + 4)\, y = x\cos x$
   (d) ⋆ $(\mathrm{D}^2 + 1)\, y = \mathrm{e}^{x}\sin x$

8. **Case 6 (general method).** Using partial fractions in D and the foundational formula $\dfrac{1}{\mathrm{D} - a} f = \mathrm{e}^{ax}\displaystyle\int \mathrm{e}^{-ax} f \,\mathrm{d}x$, find the particular integral of
$$(\mathrm{D}^2 + 1)\, y = \csc x,$$
and confirm it agrees with your answer to Question 1(b).

9. **Synthesis — solve completely.** Split each right-hand side and apply the matching case to each piece; watch for resonance.
   (a) $(\mathrm{D}^2 - 4\mathrm{D} + 3)\, y = \mathrm{e}^{3x} + x$
   (b) ⋆ $(\mathrm{D}^2 + 1)\, y = x + \sin x$

10. ⋆ **Method selection.** For each forcing below, decide whether undetermined coefficients *suffices*, or whether you must fall back on variation of parameters / the general inverse-operator case. Justify each answer in one line by the closed-family property.
    (a) $x^2\mathrm{e}^{x}$ (b) $\tan x$ (c) $x\cos x$ (d) $\dfrac{1}{x}$ (e) $\mathrm{e}^{x}\ln x$

End of Tutorial 3.